

MICROBIOLOGY – SHORT EXAM NOTES

Immune Response – Humoral (HMI) and Cell-Mediated (CMI) (Ananthanarayan & Paniker, Ch. 15)

1. INTRODUCTION

The **immune response** is the **specific reactivity induced in a host by an antigenic stimulus**. It usually protects against microorganisms but includes reactions against any antigen, living or non-living, and may be **beneficial, indifferent or injurious**. It also includes **tolerance** – specific non-reactivity. There are two types, usually developing together and acting in conjunction: **humoral (antibody-mediated) immunity** and **cell-mediated immunity**.

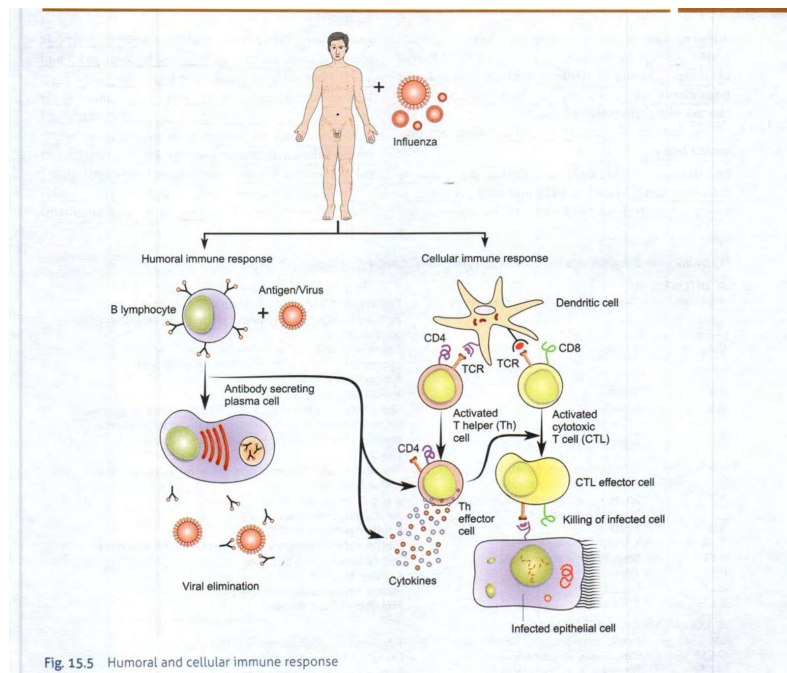


Fig. 15.5 Humoral and cellular immune response

Fig. 15.5 Humoral and cellular immune response (from the book)

HMI vs CMI – Functions

Humoral (antibody-mediated) immunity – AMI/HMI	Cell-mediated immunity – CMI
Mediated by antibodies from plasma cells (B cells)	Mediated by sensitised T lymphocytes (TH, CTL) and cytokines
Primary defence against most extracellular bacteria and toxins	Protects against intracellular pathogens – fungi, parasites, viruses, facultative intracellular bacteria (TB, leprosy)
Defence against viruses infecting via respiratory/intestinal tracts	Graft rejection and graft-versus-host reaction
Prevents recurrence of virus infections	Immunological surveillance and immunity against cancer
Pathogenesis of type 1, 2, 3 hypersensitivity and some autoimmune diseases	Pathogenesis of type 4 (delayed) hypersensitivity and some autoimmune diseases
Passive transfer by serum	Passive transfer by lymphocytes / transfer factor , not serum

2. HUMORAL IMMUNE RESPONSE (ANTIBODY-MEDIATED)

Steps in Antibody Production

1. **Afferent limb** – entry of antigen, its distribution and fate in tissues, and contact with immunocompetent cells.
2. **Central functions** – processing of antigen by cells and control of antibody formation.
3. **Efferent limb** – secretion of antibody, its distribution in tissues and body fluids, and its effects.

Phases of Antibody Response

1. **Lag phase** – no antibody detectable; 2. **Log phase** – titre rises steadily; 3. **Plateau (steady state)** – synthesis equals catabolism; 4. **Phase of decline** – catabolism exceeds production and titre falls.

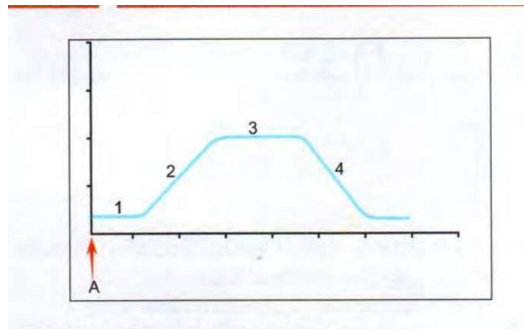


Fig. 15.1 Primary immune response – A: antigen; 1 lag, 2 log, 3 plateau, 4 decline (from the book)

Primary and Secondary Responses

The **primary response** is the antibody response to the first antigenic stimulus; the **secondary (anamnestic/booster) response** follows later contact with the same antigen and is faster, stronger and longer-lasting because of **memory cells**.

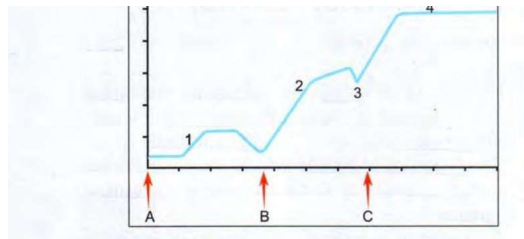


Fig. 15.2 Effect of repeated antigenic stimulus – A, B, C: successive doses of antigen (from the book)

Feature	Primary response	Secondary response
Lag period	Long – usually 4–7 days (2–3 weeks with diphtheria toxoid)	Short – usually 1–3 days
B cell involved	Naïve B cell	Memory B cell
Peak response	7–10 days	3–5 days
Magnitude	Low; depends on antigen	100–1000 times higher
Antibody class	IgM predominates early	IgG predominates
Antigens	T-dependent and T-independent	Only T-dependent
Antibody affinity	Lower	Higher
Immunisation	High dose; better with adjuvants	Adjuvants not required

Priming and booster doses: a single dose of a non-living vaccine only primes; protective levels need **booster doses**. A **live vaccine** needs one dose, as multiplication acts as both priming and booster. **Negative phase:** injecting antigen into an animal already having antibody causes a temporary fall in antibody level (antigen combines with antibody), followed by a rise.

Fate of Antigen in Tissues

Intravenous antigens localise in **spleen, liver, bone marrow, kidneys and lungs**; 70–80% eliminated in 1–2 days. Subcutaneous antigens localise in **draining lymph nodes**. **Particulate antigens** are removed in two phases (non-immune phagocytosis, then **immune elimination**); **soluble antigens** in three phases – **equilibration, metabolism and immune elimination** (antigen–antibody complexes may cause **serum sickness**). Proteins are eliminated in days to weeks; **polysaccharides persist** for months or years (pneumococcal polysaccharide up to 20 years).

Mechanism of Antibody Production

Three cell types take part: **antigen-presenting cells (macrophages, dendritic cells), T cells and B cells**.

- **Antigen processing and presentation** – APCs capture and process antigen and present it with **MHC** to T cells. Needed for **T-dependent antigens** (proteins, RBCs); **T-independent antigens** (polysaccharides with repeating epitopes) stimulate B cells directly.
- **TH cell activation** – needs **two signals**: (1) TCR binding to antigen–**MHC class II**, (2) **IL-1** from the APC. Activated TH cells secrete **IL-2** and **IL-4, IL-5, IL-6** (B cell growth and differentiation factors).
- **B cell activation** – B cells that have bound antigen **proliferate clonally** and differentiate into **plasma cells** secreting antibody; a few become long-lived **memory B cells** (secondary response).
- **Cytotoxic T cells** – activated by antigen with **MHC class I** plus **IL-2** from TH cells; kill virus-infected or tumour cells; some become memory cells.

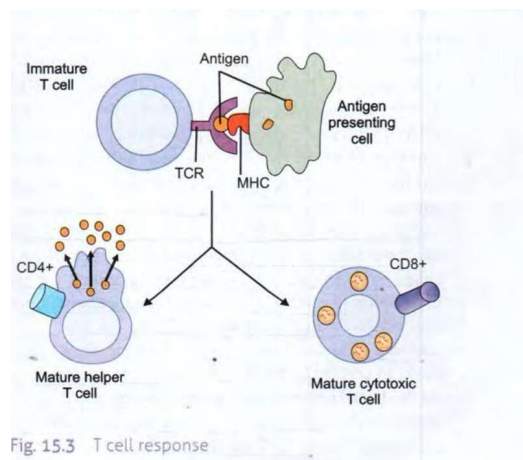


Fig. 15.3 T cell response (from the book)

Factors Influencing Antibody Production

Factor	Effect
Genetic	Responders and non-responders; controlled by Ir (immune response) genes
Age	Embryo immature; infant depends on maternal antibody till 3–6 months; full competence – IgG by 5–7 yrs, IgA by 10–15 yrs
Nutrition	Protein–calorie malnutrition suppresses HMI and CMI (CMI more); amino acid and vitamin deficiencies reduce antibody

Factor	Effect
Route	Parenteral better than oral/nasal; oral/nasal favours IgA ; inhaled pollen – IgE ; deltoid better than gluteal for hepatitis B vaccine
Dose	Effective above a minimum dose; excess dose may cause tolerance – immunological paralysis (Felton, 1949)
Multiple antigens	Response may be unchanged, enhanced (pertussis enhances toxoids in DPT) or diminished (antigenic competition)
Adjuvants	Enhance immunogenicity – aluminium hydroxide/phosphate (used in vaccines), Freund's incomplete (water-in-oil) and complete (+ killed tubercle bacilli; granuloma, not for humans), muramyl dipeptide , LPS, <i>B. pertussis</i>
Immunosuppressive agents	Irradiation, radiomimetic drugs (cyclophosphamide) , corticosteroids , antimetabolites (methotrexate, azathioprine, 6-MP), cyclosporine (inhibits TH cells; most widely used), anti-lymphocyte serum
Effect of antibody	Passive antibody suppresses response (feedback) – basis of anti-Rh globulin within 72 h of delivery; give toxoid and antitoxin at separate sites

Monoclonal Antibodies

Antibodies produced by a **single clone** against a **single antigenic determinant** (as in multiple myeloma). **Hybridoma technology – Köhler and Milstein (1975; Nobel 1984)**: spleen cells from an immunised mouse are fused with **myeloma cells** lacking the enzyme **HGPRT**; hybrids are selected in **HAT medium** (hypoxanthine, aminopterin, thymidine), where unfused myeloma cells die and normal lymphocytes cannot grow indefinitely. Antibody-producing hybridomas are cloned, grown in culture or as ascites in mice, and can be frozen. **Humanised (chimeric)** antibodies reduce anti-mouse reactions. **Uses**: diagnosis (ELISA, immunofluorescence kits), therapy and research.

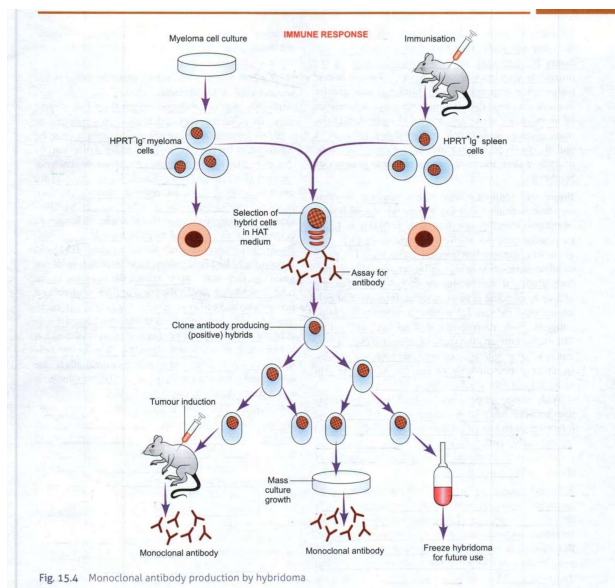


Fig. 15.4 Monoclonal antibody production by hybridoma (from the book)

3. CELL-MEDIATED IMMUNE RESPONSE

CMI refers to specific immune responses **mediated by T cells** (TH cells and **cytotoxic T lymphocytes** as effectors) that **do not involve antibodies**. Its classic demonstration is **delayed hypersensitivity (DH)** – e.g. the **tuberculin test** (Koch, 1890): an indurated nodule with mononuclear infiltration appearing **48–72 hours**

after intradermal antigen. It **cannot be transferred by serum**, but was transferred by **leucocytes** (Landsteiner and Chase, 1942).

Scope of CMI

- **Delayed hypersensitivity.**
- Immunity to **intracellular parasites** – bacteria (**tuberculosis, leprosy**, listeriosis, brucellosis), fungi (histoplasmosis, coccidioidomycosis), protozoa (leishmaniasis, trypanosomiasis) and viruses (measles, mumps).
- **Transplantation immunity** and graft-versus-host reaction.
- **Immunological surveillance** and immunity against **cancer**.
- Pathogenesis of some **autoimmune diseases** (thyroiditis, encephalomyelitis).

Induction of CMI

Best induced by infection with **intracellular parasites**; killed vaccines and non-living antigens do not induce CMI unless given with **Freund-type adjuvants**; only **T-dependent antigens** induce it; some chemicals on skin cause DH (contact dermatitis). Each T cell has a **TCR** for one epitope; on meeting antigen with MHC it undergoes **blast transformation, clonal proliferation** and differentiation into **effector and memory cells**. **Helper T cells** recognise antigen with **MHC class II** on macrophages and release **cytokines** that **activate macrophages** to kill intracellular parasites. **Cytotoxic T cells** recognise antigen with **MHC class I** on virus-infected, tumour or graft cells and **destroy them**.

Cytokines

Cytokines are peptide mediators (intercellular messengers) that regulate immunological, inflammatory and reparative responses; highly potent (active at femtomolar concentrations); produced by widely distributed cells; act **locally (paracrine/autocrine)**; **pleiotropic** with overlapping effects. Products of lymphocytes were called **lymphokines**, of monocytes **monokines**; those regulating other leucocytes are **interleukins**.

Cytokine	Main source	Key functions
IL-1	Macrophages	Activates T cells (IL-2 production); endogenous pyrogen ; acute phase proteins; with TNF causes changes of septic shock
IL-2 (T cell growth factor)	T cells	Major activator of T and B cells; stimulates CTL and NK; forms LAK cells
IL-3 (multi-CSF)	T cells	Growth of bone marrow stem cells
IL-4	TH2 cells	B cell activation; IgE synthesis (atopy); T and mast cell growth
IL-5	TH2 cells	B cell proliferation; eosinophil maturation; IgA
IL-6	T, B, macrophages	Immunoglobulin synthesis; acute phase proteins
IL-8	Macrophages	Neutrophil chemotaxis
IL-12	T cells/macrophages	Activates NK cells
IL-17	TH17 cells	Proinflammatory; autoimmunity
TNF-α (cachectin) / TNF-β (lymphotoxin)	Macrophages / T cells	Tumour necrosis, cachexia, endotoxic shock , phagocyte activation
IFN-α, β, γ	Leucocytes, fibroblasts, T cells	Antiviral ; IFN-γ activates macrophages and increases MHC expression

Cytokine	Main source	Key functions
CSFs (GM, G, M)	T cells, fibroblasts	Growth of blood cell colonies
TGF-β	T and B cells	Wound healing; down-regulates immune responses

Classical lymphokines (Table 15.2)

Affecting macrophages – **migration inhibiting factor (MIF)**, macrophage activating factor, chemotactic factor; affecting lymphocytes – blastogenic factor, T and B cell growth factors; affecting granulocytes – chemotactic factor, CSF; affecting cultured cells – **lymphotoxin, interferon, TNF**; others – skin reactive factor, **transfer factor**.

Detection of CMI

- **Skin test for delayed hypersensitivity** – e.g. **tuberculin (Mantoux) test** (original method).
- **In vitro: lymphocyte transformation test** (blast transformation of sensitised T cells with antigen), **target cell destruction** (cytotoxicity), and **migration inhibition factor (MIF) test** – migration of macrophages/leucocytes from a capillary tube is **inhibited** when the specific antigen is added.

Transfer Factor (TF)

Described by **Lawrence (1954)** – an extract of leucocytes from sensitised donors that **transfers CMI** (not humoral immunity) to all antigens to which the donor is sensitive; transfer is **systemic** and can be serial. TF is a **dialysable, low-molecular-weight (2000–4000)** substance, resistant to trypsin, DNase and RNase, **inactivated at 56°C for 30 min, non-antigenic**; probably a polypeptide–polynucleotide. **Uses:** T cell deficiency (**Wiskott–Aldrich syndrome**), disseminated infections with deficient CMI (lepromatous leprosy, TB, mucocutaneous candidiasis), malignant melanoma.

Summary comparison – HMI vs CMI

Feature	HMI	CMI
Cells	B cells → plasma cells	T cells (TH, CTL), macrophages
Effector	Antibodies	Sensitised T cells, cytokines
MHC	Class II (TH help)	Class I (CTL) and class II (TH)
Transfer	Serum	Leucocytes / transfer factor
Hypersensitivity	Types I, II, III (immediate)	Type IV (delayed)
Tests	Serology (agglutination, ELISA)	Tuberculin test, LTT, MIF test

4. IMMUNOLOGICAL TOLERANCE AND THEORIES (IN BRIEF)

Immunological tolerance is the condition in which contact with an antigen **specifically abolishes the capacity to mount an immune response** to that antigen later; response to other antigens is unaffected. **Burnet and Fenner (1949)** suggested that self-tolerance results from contact with self-antigens during embryonic life; **Medawar (1953)** proved it – mice injected with CBA cells as newborns later accepted CBA skin grafts. Tolerance depends on **species and immunocompetence** (easiest in embryos/newborns), **nature of antigen** (soluble more tolerogenic), **dose (high-zone and low-zone tolerance)** and **route** (intravenous). High doses induce **B cell** tolerance; repeated minute doses induce **T cell** tolerance. **Split tolerance** – unresponsiveness in one arm only.

Theories of antibody formation: instructive – **Ehrlich's side chain theory (1900)**, **direct template** (Breinl & Haurowitz; Pauling), **indirect template** (Burnet & Fenner); selective – **natural selection (Jerne, 1955)** and

clonal selection (Burnet, 1957) – the currently accepted theory: each lymphocyte clone reacts with one antigen; self-reactive '**forbidden clones**' are eliminated in embryonic life; antigen selects and expands the specific clone. **Jerne's network (idiotype) hypothesis** explains regulation; **Tonegawa** (1987 Nobel) explained antibody diversity by split genes.

★ **EXAM FOCUS**

- Short answers: **Types of immune response and their functions, Monoclonal antibodies – principle and preparation.**
- Short notes: **Humoral immunity, CMI, primary vs secondary response, factors influencing antibody production, cytokines, transfer factor, immunosuppressive agents, immunological tolerance, theories of immune response.**
- One-liners: early antibody – IgM; secondary – IgG; TH activation – TCR + IL-1; hybridoma – HAT medium; DH appears at 48–72 h; clonal selection – Burnet.